

The implementation plan is approved with the following modifications and decisions:

Approved Decisions

1. Proceed with fixing all critical bugs and missing APIs in Phase 1.
2. Keep Telegram Bot integration for worker notifications during development.
3. Add Firebase FCM, Email, and SMS notifications in later phases using BullMQ + Redis.
4. Continue using sandbox fallbacks for Razorpay, Maps, and AI services when credentials are unavailable, while implementing production-ready code paths.

Architecture Decisions

Next.js Version

Do NOT start with an App Router migration.

Keep the current Pages Router until:

- Critical bugs are fixed
- Backend refactoring is complete
- Test coverage is established
- Security hardening is complete

App Router migration should be moved to a later phase after the system becomes stable.

SMS Gateway

Use MSG91 as the primary SMS provider because the platform targets Indian users. Keep the SMS layer provider-agnostic so Twilio can be added later if required.

AI Chatbot Knowledge Base

Create an initial structured knowledge base:

backend/data/

- faq.json
- policies.json
- services.json
- pricing.json

Build the chatbot on top of these files first.

Do not introduce Pinecone, Weaviate, or other vector databases in the initial implementation.

Additional Mandatory Phase

Add a new phase before Socket.IO upgrades:

Security Hardening Phase

Implement:

- Helmet
- Rate Limiting
- Refresh Tokens
- Token Rotation
- Request Validation
- XSS Protection
- CSRF Protection
- Audit Logging
- Secure Cookie Configuration
- Environment Variable Validation

Security improvements must be completed before AI features.

Additional Requirements

Git Workflow

Establish:

- main branch
- develop branch
- feature branches
- protected main branch
- pull request workflow

Environment Management

Create:

- .env.example
- .env.development
- .env.staging
- .env.production

Validate all environment variables during application startup.

Database Migration Strategy

Introduce migration tooling so schema changes can be applied safely in production environments.

Revised Execution Order

Phase 0: Git Workflow + Environment Strategy + Database Migrations

Phase 1: Critical Bug Fixes & Missing APIs

Phase 2: Database Improvements + Booking State Machine

Phase 3: Clean Architecture Refactor

Phase 4: Security Hardening

Phase 5: Socket.IO & BullMQ Upgrades

Phase 6: Frontend Improvements

Phase 7: Testing + Observability

Phase 8: DevOps + CI/CD

Phase 9: AI Features

Phase 10: Optional App Router Migration

Before starting implementation, provide a detailed file-by-file execution plan for Phase 0 and Phase 1 and wait for approval.